Classification for Device States

Classification turns sensor data into actionable decisions

What is Classification?

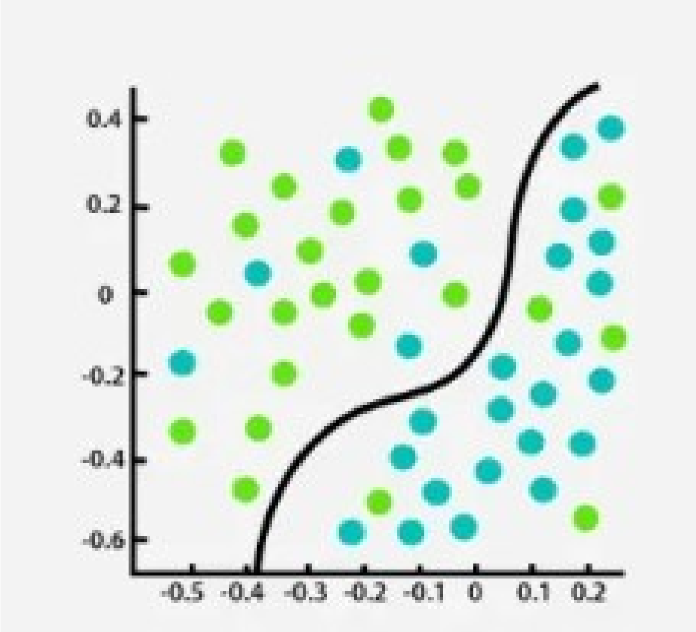
🏷 Predicts categories or classes
✓ Output is a label, not a number

AIoT Applications

⚙ Machine state detection (Normal • Warning • Failure)
🚨 Intrusion & security event detection
🔧 Faulty sensor identification
🔄 Operating mode classification

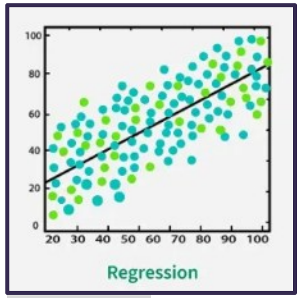
Why Classification Matters

⚡ Enables real-time decisions
🤖 Automates monitoring & alerts
🛡 Improvs safety & reliability



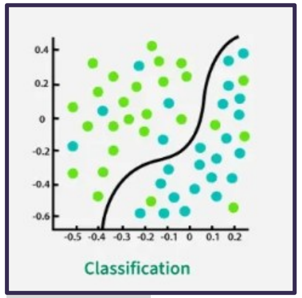
Common Algorithms
✓ Logistic Regression
✓ Decision Trees
✓ Random Forest
✓ Gradient Boosting

Classification vs Regression



Regression

- ✓ Predicts **continuous numerical values**
  ✓ Answers: *How much? How long? How many?*
- **Examples**
  🌡 Predict temperature
  ⚡ Forecast energy consumption
  ⏳ Estimate remaining equipment life



Classification

- ✓ Predicts **categories or labels**
  ✓ Answers: *What state? What type? What event?*
- **Examples**
  ⚙ Machine state: Normal / Warning / Failure
  🚨 Intruder detected vs not detected
  🔧 Faulty vs healthy sensor

Regression predicts values — Classification predicts decisions

Key Classification Techniques in AIoT

Technique selection depends on data complexity, accuracy needs, and device constraints

Deep Learning (DL)

- ✓ Handles complex data: images, video, audio
  ✓ CNNs for object & pattern recognition
- **Use:** smart surveillance, defect detection, voice commands

Machine Learning (ML)

- ✓ Decision Trees
  ✓ Random Forest
  ✓ Support Vector Machines (SVM)
  ✓ K-Nearest Neighbours (KNN)
- **Use:** device state detection, anomaly classification, fault detection

TinyML Models

- ✓ Optimized for low-power devices
  ✓ Runs directly on sensors & microcontrollers
- **Use:** wearable alerts, on-device fault detection, smart sensors

Hybrid Models

- ✓ Combine multiple techniques (e.g., CNN + BiLSTM)
  ✓ Improve accuracy & fault diagnosis
- **Use:** predictive maintenance, complex industrial monitoring

Primary Areas of AIoT Classification Analysis

🏭 Industrial Fault Classification

- ✓ Analyse time-series sensor data
  ✓ Identify specific machine failure conditions
  ✓ Often uses hybrid deep learning models
- **Impact:** Prevent downtime & improve safety

♻️ Smart Waste Classification

- ✓ CNN-based image classification
  ✓ Edge devices (e.g., Jetson Nano) sort waste
  - **Classifies:** plastic • paper • metal • organic
    Impact: Improves recycling & smart city efficiency

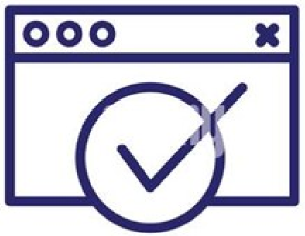
🔐 IoT Security & Attack Detection

- ✓ Analyze network traffic patterns
  ✓ Detect cyber threats & anomalies
- **Example:** Random Forest detects DDoS attacks (>99% accuracy)
  Impact: Protects connected infrastructure

🏃 Activity Recognition

- ✓ Classify human activities using sensor data
  ✓ Wearables & smart environments
- **Detects:** walking • sleeping • eating • falls
  Impact: Enables healthcare & assisted living solutions

Key Performance Evaluation Metrics



Accuracy

- Measures overall correctness of predictions
- **Example:** % of correctly classified machine states

Precision

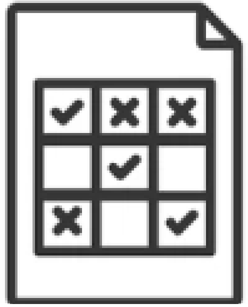
- Measures how many predicted positives are correct
- **Important for:** avoiding false alarms
- **Example:** security attack alerts

Recall

- Measures how many actual positives are detected
- **Important for:** catching critical events
- **Example:** detecting rare failures

F1-Score

- Balances precision & recall
- Useful for imbalanced datasets
- **Example:** intrusion detection systems



Confusion Matrix

- Visualizes prediction performance
- Shows correct vs incorrect classifications
- Identifies misclassification patterns

Challenges in AIoT Classification

Resource Constraints

✓ Limited CPU, memory & storage
✓ Edge devices cannot run heavy models

Impact: requires lightweight & optimized AI

Data Scarity & Quality Issues

✓ Limited labeled datasets
✓ Rare events are hard to capture
✓ Noisy sensor data reduces accuracy

Impact: affects model reliability

Non-IID Data (Data Bias)

✓ Data varies by device, location & environment
✓ Patterns differ across deployments

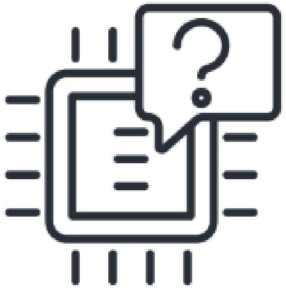
Impact: models may perform poorly in new conditions

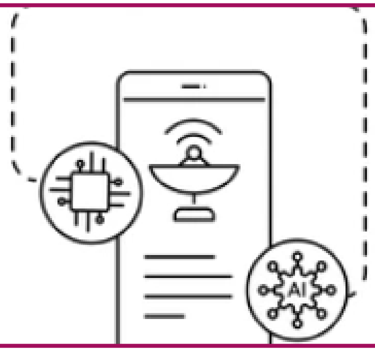
Security & Privacy Concerns

✓ Sensitive data processed at the edge
✓ Risk of data leakage or misuse

Impact: requires secure & privacy-preserving AI

Future Direction





🔍 Explainable & Interpretable AI

✓ Make model decisions transparent
✓ Improve trust & regulatory compliance

⚡ Efficient Edge AI Algorithms

✓ Ultra-lightweight models for low-power devices
✓ Faster inference with minimal resources

📡 Robust Performance in Dynamic Environments

✓ Handle noisy & changing sensor data
✓ Adapt to new conditions in real time

🧠 Adaptive & Self-Learning Systems

✓ Continuous learning from live data
✓ Improved accuracy over time

Future Direction

🏷 Role of Classification in AIoT

- ✓ Detect device states & operating modes
  ✓ Identify faults, anomalies & security threats
  ✓ Enable real-time automated decisions

🧠 Techniques & Approaches

- ✓ Machine Learning (DT, RF, SVM)
  ✓ Deep Learning for complex data
  ✓ TinyML for edge intelligence
  ✓ Hybrid models for improved accuracy

📊 Ensuring Reliable Performance

- ✓ Use accuracy, precision, recall & F1-score
  ✓ Analyze confusion matrix for improvements

⚠ Challenges & Future Direction

- ✓ Resource & data constraints
  ✓ Privacy & security requirements
  ✓ Need for explainable & adaptive AI